CREDIT RISK & LOAN DEFAULT ANALYSIS

A Data Analytics Case Study

Dataset: Credit Risk | Source: Kaggle

Python, Pandas, Data Visualization, MySQL

Which borrower segments carry the highest default risk, and what financial signals best predict it?

1. Background & Business Problem

Lending is a core function of any bank or financial institution. When a bank gives someone a loan, it is making a bet that the borrower will repay it. Sometimes that bet goes wrong — the borrower stops paying, and the bank loses money. In the industry, this is called a loan default, and reducing it is one of the most important goals in credit risk management.

The problem is not just about identifying defaulters after the fact. The real value is in identifying who is likely to default before a loan is approved or before early warning signs are missed. Even a small reduction in default rates can save financial institution millions every year.

This project looks at a real-world dataset of approximately 150,000 borrowers from an American lending institution. Each borrower has financial attributes recorded — their monthly income, how much of their credit limit they are using, how many times they have been late on payments, their age, number of dependents, and most importantly, whether they defaulted on a loan within two years.

Business Question: Which types of borrowers are most likely to default on a loan, and what financial characteristics are the strongest predictors of credit default?

2. About the Dataset

The dataset used in this project is the "Give Me Some Credit" dataset, publicly available on Kaggle. It contains records of 150,000 borrowers with the following key features:

Column Name	What It Represents
SeriousDlqin2yrs	Target variable — 1 if borrower defaulted, 0 if not
RevolvingUtilizationOfUnsecuredLines	Credit utilization ratio (credit used / credit limit)
age	Age of the borrower in years
DebtRatio	Monthly debt payments divided by monthly income
MonthlyIncome	Monthly income of the borrower (in USD)
NumberOfTime30-59DaysPastDueNotWorse	Times the borrower was 30–59 days late on a payment
NumberOfTimes90DaysLate	Times the borrower was 90+ days late
NumberOfOpenCreditLinesAndLoans	Total number of open credit accounts and loans
NumberOfDependents	Number of people financially dependent on the borrower

The dataset had a mix of clean and messy data — which made it a realistic and useful exercise.

3. Data Cleaning

Before doing any analysis, I spent time understanding the problems in the raw data. This step is the most important one.

3.1 Problems Found in the Raw Data

- MonthlyIncome had **29,731 missing values** — nearly 20% of all rows had no income recorded.
- NumberOfDependents had **3,924 missing values** — smaller but still needed handling.
- RevolvingUtilizationOfUnsecuredLines had a maximum value of **50,708** — this is a ratio that should be between 0 and 1.
- DebtRatio had a maximum value of **329,664** — another ratio column with impossible values caused by bad data entry.
- Age had a **minimum value of 0** — no bank lends to a newborn. These rows were data entry errors.
- MonthlyIncome had extreme outliers at the top — values up to **$3 million per month** that would distort all calculations.
- An **unnamed index column** was auto-saved into the CSV file and needed to be removed.

3.2 How Each Problem Was Fixed

Problem	Fix Applied	Reason
Missing MonthlyIncome (29,731 rows)	Filled with median income ($5,400)	Median is more robust than mean for skewed income data
Missing NumberOfDependents (3,924 rows)	Filled with 0	Absence of dependents data most likely means zero dependents
Age = 0 (impossible values)	Removed those rows entirely	No bank lends to a 0-year-old. These are errors.
DebtRatio outliers (max: 329,664)	Capped at 10	Even the most extreme debt burden wouldn't exceed 10x income
RevolvingUtilization outliers (max: 50,708)	Capped at 1.0	This is a ratio — anything above 1 is over-limit, not 50,000
MonthlyIncome extreme outliers	Capped at 99th percentile ($23,000)	Preserves distribution shape while removing impossible extremes
Unnamed index column	Dropped	Not a real feature — just a CSV artifact

After cleaning: 149,986 rows remained with 11 columns. Zero missing values. All columns within realistic ranges. The data was now ready for analysis.

4. Feature Engineering

Raw numbers like age = 34 or income = 4200 are hard to analyze in bulk. To find patterns, you need to group similar borrowers together.

Four new grouping columns were created:

New Feature	Groups Created	Why This Grouping
AgeBand	21-30, 31-40, 41-50, 51-60, 61-70, 71+	Life stages have very different financial behaviors
IncomeBand	Low (<$3K), Medium ($3K-$6K), High ($6K-$10K), Very High (>$10K)	Income level is a core driver of repayment capacity
UtilizationGroup	Low (0-30%), Medium (30-60%), High (60-80%), Very High (80-100%)	Credit utilization measures financial stress levels
DebtRatioGroup	Low (<0.25), Moderate (0.25-0.5), High (0.5-1.0), Very High (>1.0)	Debt-to-income is a classic credit risk signal

Additionally, a simple Rule-Based Risk Score was built by assigning 1 point for each of the following risk flags present in a borrower's profile:

- Credit utilization above 80%
- Debt ratio above 0.5
- Any history of being 90+ days late on a payment
- Any history of being 30-59 days late on a payment
- Monthly income below $3,000

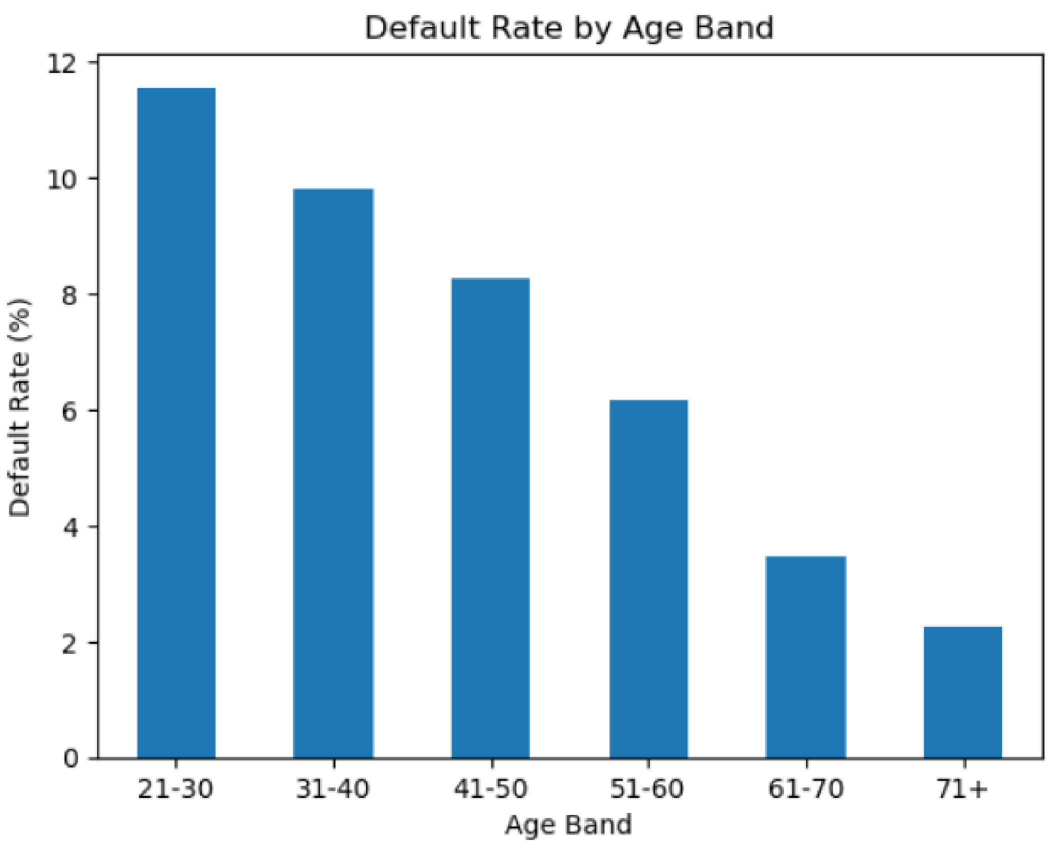
This gave each borrower a score from 0 (no risk flags) to 5 (all risk flags triggered). The purpose was to validate our analysis — if higher scores strongly correlate with higher default rates, our understanding of the risk drivers is correct.

5. Key Findings — What the Data Revealed

The overall default rate in this dataset is 6.68%. This means roughly 1 in every 15 borrowers defaulted on their loan within two years. Every segment-level finding below is measured against this baseline.

5.1 Default Rate by Age

The analysis shows a clear and consistent pattern — younger borrowers default at significantly higher rates than older borrowers.

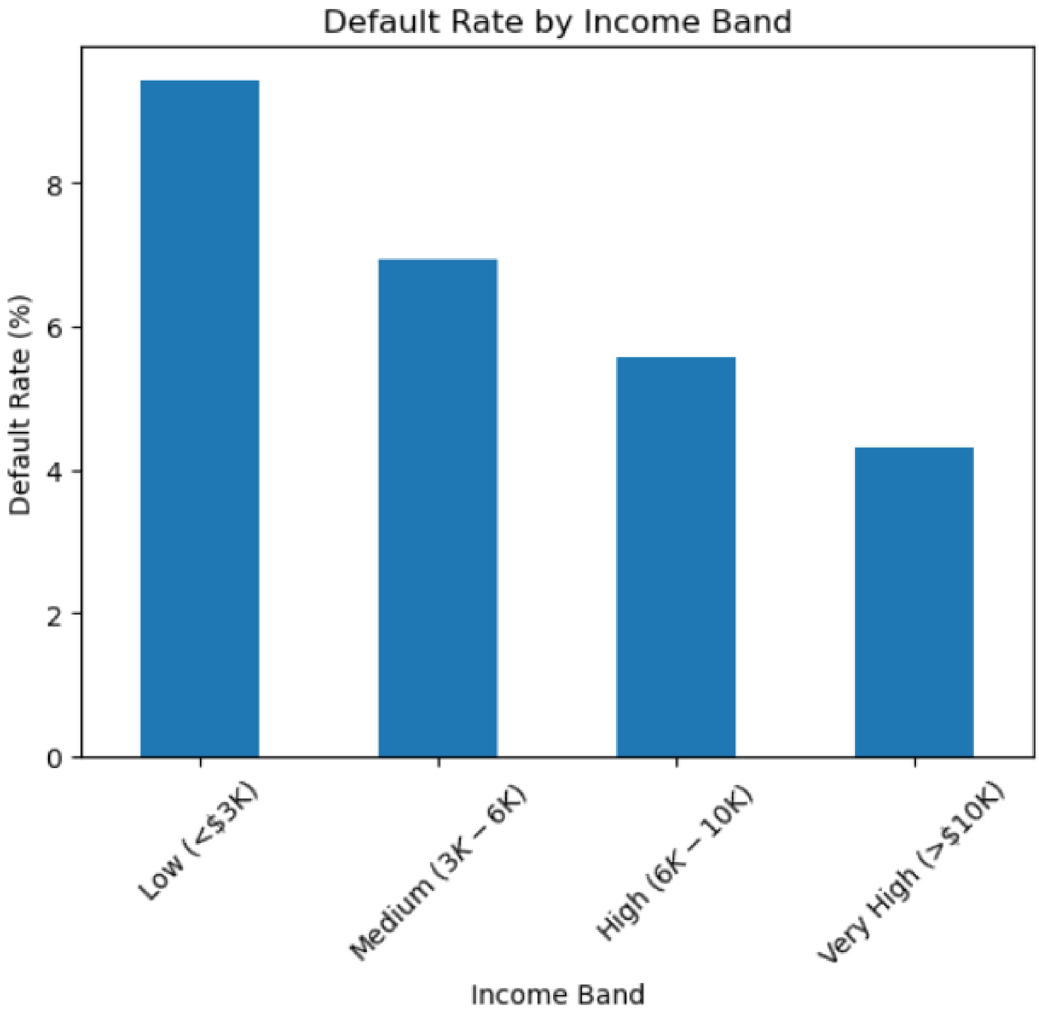


Age Band	Total Borrowers	Defaulters	Default Rate
21-30	10,757	1,244	11.56%
31-40	24,339	2,390	9.82%
41-50	35,037	2,893	8.26%
51-60	34,806	2,149	6.17%
61-70	27,424	952	3.47%
71+	17,623	397	2.25%

Key Insight: Borrowers aged 21-30 default at 11.56% — nearly 1.7 times the overall average. Borrowers aged 71+ default at only 2.25%, which is less than one-third of the overall rate. Age is a meaningful signal of credit risk, likely because younger borrowers have less financial stability and savings.

5.2 Default Rate by Monthly Income

There is a clear inverse relationship between income and default risk. As income increases, default rate falls consistently. This makes intuitive sense — higher income means more capacity to repay debts even when unexpected expenses arise.

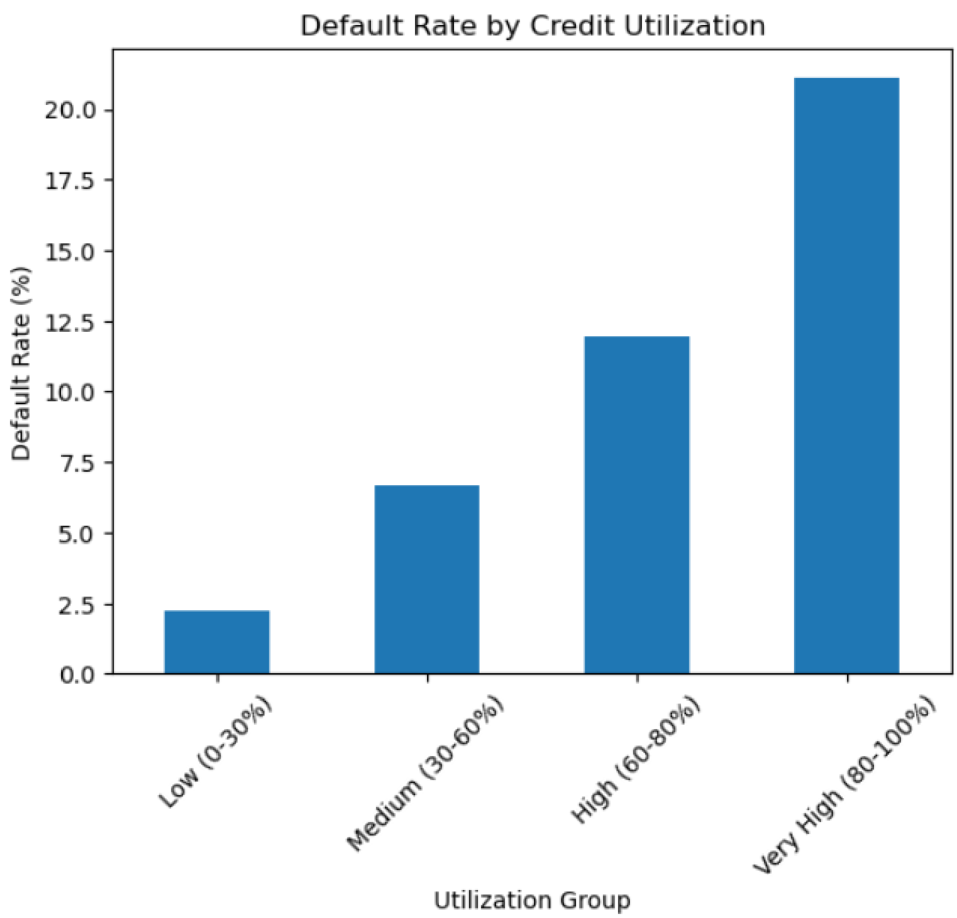


Income Band	Total Borrowers	Defaulters	Default Rate
Low (<$3K/month)	23,446	2,210	9.43%
Medium ($3K-$6K/month)	73,711	5,123	6.95%
High ($6K-$10K/month)	32,876	1,834	5.58%
Very High (>$10K/month)	18,319	792	4.32%

Key Insight: Low income borrowers default at 9.43% — more than twice the rate of very high income borrowers (4.32%). Income is a fundamental determinant of repayment capacity. The larger the income cushion, the more resilient a borrower is to financial shocks.

5.3 Default Rate by Credit Utilization — The Strongest Signal

Credit utilization is how much of your available credit limit you are actively using. A person who has a $10,000 credit limit and is using $8,500 has a utilization of 85% — they are close to maxed out. This is the single strongest predictor of default in the entire dataset.

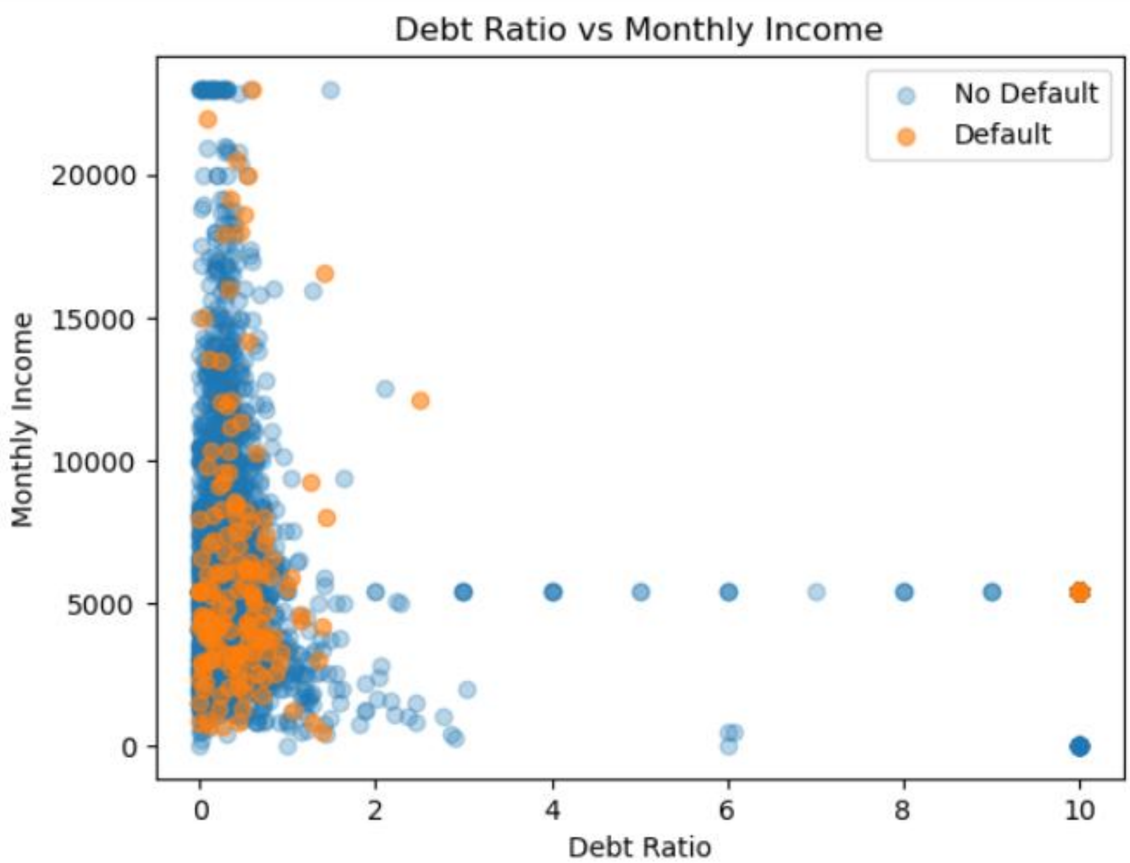


Utilization Group	Total Borrowers	Defaulters	Default Rate
Low (0-30%)	92,870	2,060	2.22%
Medium (30-60%)	21,887	1,462	6.68%
High (60-80%)	10,098	1,205	11.93%
Very High (80-100%)	25,131	5,298	21.08%

Key Insight: Borrowers with credit utilization above 80% default at 21.08% — nearly 9.5 times the rate of low-utilization borrowers (2.22%). High utilization signals financial stress — the borrower is living close to or at their credit limit, leaving no buffer for unexpected expenses.

5.4 Debt Ratio vs Monthly Income

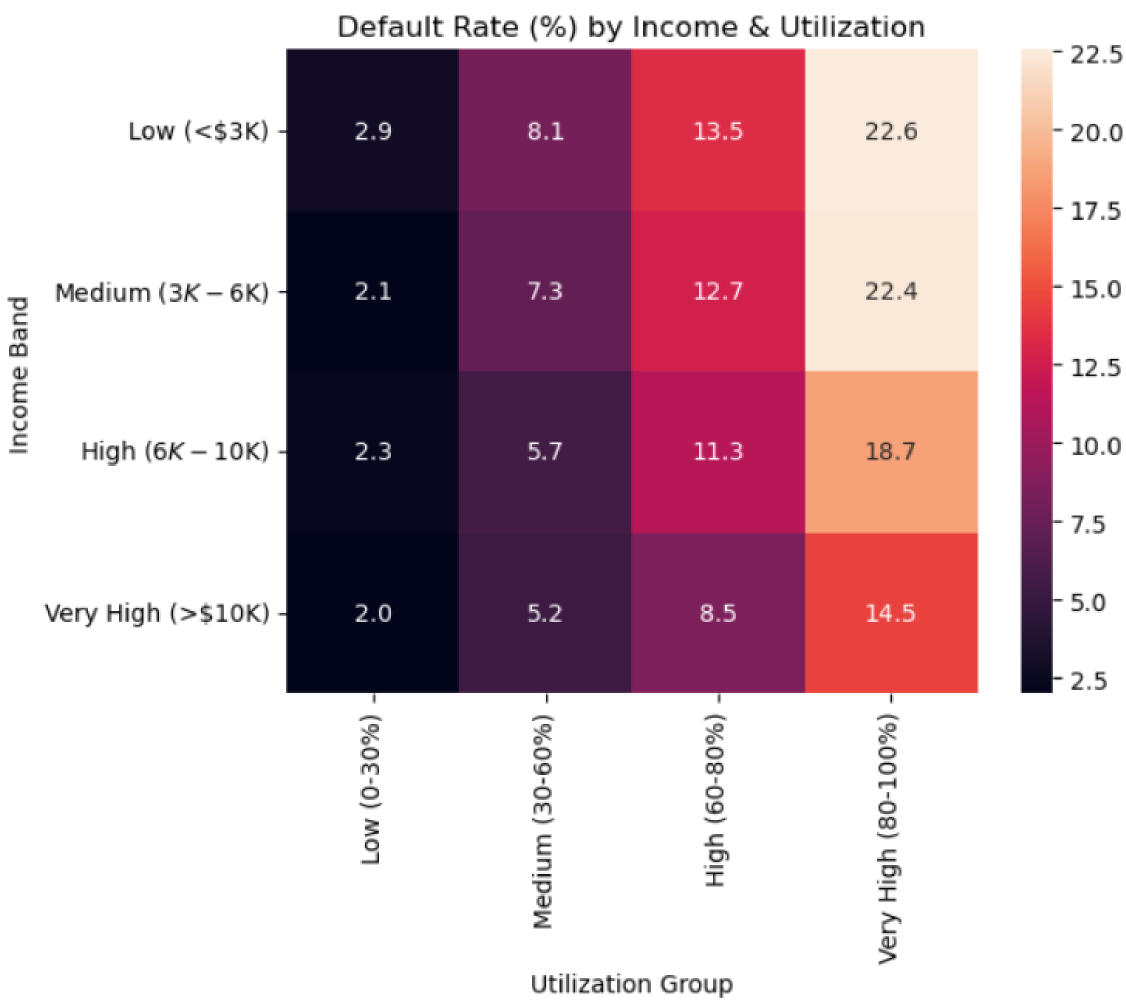
The scatter plot below shows the relationship between a borrower's debt ratio and their monthly income, with defaulters highlighted in orange. It visually confirms that defaults are concentrated at low income levels regardless of debt ratio, and that high debt ratios (above 2.0) are associated with both low income and default.



The chart also reveals something interesting — there is a visible horizontal band of points at $5,400 income. This is the median that was used to fill missing income values during cleaning. This is a common artifact of median imputation and is worth noting as a data limitation.

5.5 Combined Effect — Income + Utilization Heatmap

Looking at one variable at a time is useful, but the most powerful insights come from combining variables. The heatmap below shows default rates when income band and credit utilization group are considered together. The darker the red, the higher the default rate.



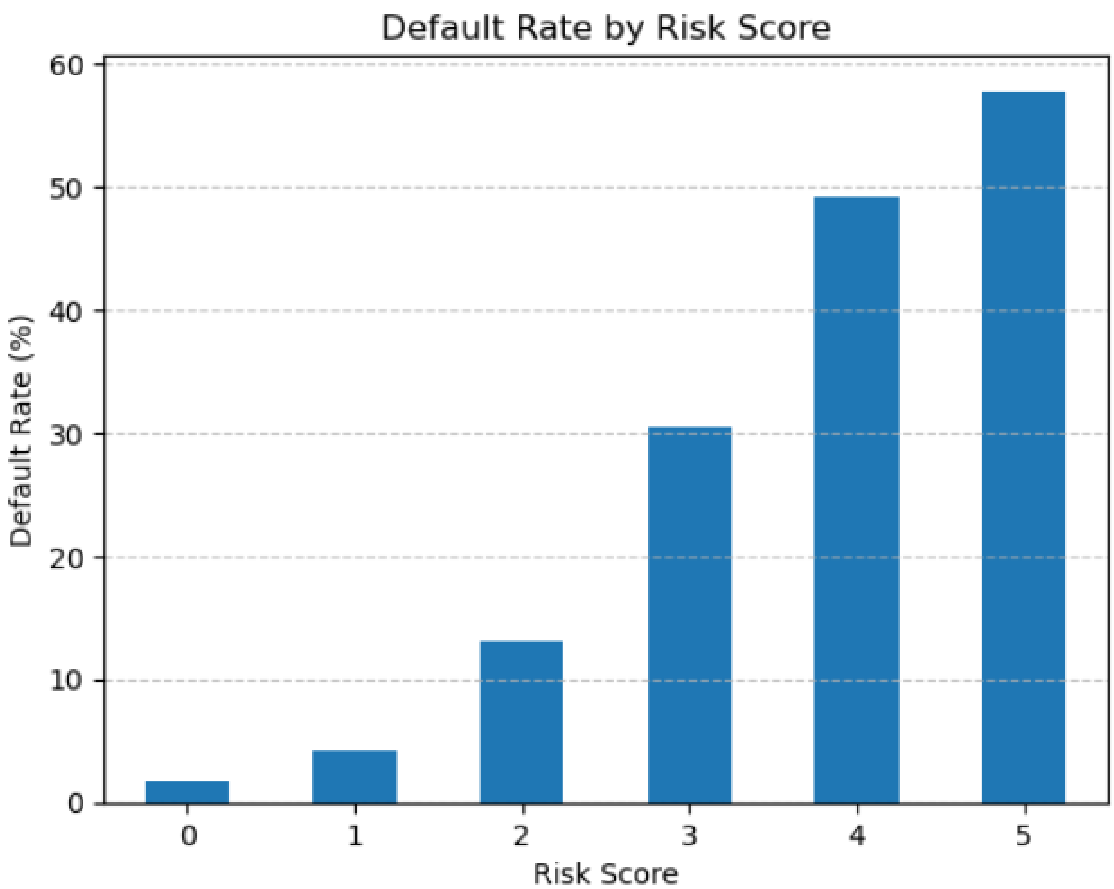
Income Band	Utilization Group	Default Rate	vs. Overall
Low (<$3K)	Very High (80-100%)	22.56%	3.38x higher
Medium ($3K-$6K)	Very High (80-100%)	22.38%	3.35x higher
High ($6K-$10K)	Very High (80-100%)	18.69%	2.80x higher

Low (<$3K)	Low (0-30%)	2.91%	0.44x — below average
Very High (>$10K)	Low (0-30%)	2.01%	0.30x — well below average

Finding: Low income borrowers with very high credit utilization default at 22.56% — 3.38 times the overall portfolio average. In contrast, high-income borrowers with low utilization default at just 2.01%. The combination of low income and maxed-out credit is the single most dangerous risk profile in this dataset.

5.6 Risk Score Validation

The rule-based risk score assigned to each borrower was validated by checking whether it actually predicted defaults. If the score is meaningful, higher scores should result in much higher default rates.



Risk Score	Total Borrowers	Defaulters	Default Rate	vs. Overall Average
0 (No risk flags)	58,750	1,022	1.74%	0.26x
1	57,796	2,474	4.28%	0.64x
2	23,491	3,065	13.05%	1.95x
3	7,757	2,367	30.51%	4.56x
4	1,953	959	49.10%	7.35x
5 (All risk flags)	239	138	57.74%	8.64x

Key Insights: The risk score shows a near-perfect gradient. Score 0 borrowers default at just 1.74%, while Score 5 borrowers default at 57.74% — 8.64 times the overall average. This confirms that the five risk factors selected (utilization, debt ratio, late payments, income) are genuinely predictive of default behavior.

6. SQL Analysis

In addition to the Python analysis in Jupyter Notebook, the cleaned dataset was loaded into a MySQL database and queried using standard SQL. This was done to demonstrate that the same analytical insights can be derived using SQL.

The database was created locally using MySQL Community Server, and the data was loaded using Python's SQLAlchemy library. Five SQL queries were written to replicate and extend the findings from the exploratory analysis.

Query	What It Does	SQL Concepts Used
Query 1	Overall default rate across all 149,986 borrowers	COUNT, SUM, AVG, ROUND
Query 2	Default rate broken down by age band	GROUP BY, ORDER BY, aggregations
Query 3	Top 20 highest risk individual borrowers	ORDER BY DESC, LIMIT
Query 4	Default rate by income + utilization combined segment	Multi-column GROUP BY, ORDER BY
Query 5	Default rate by risk score with ranking	RANK() window function, GROUP BY

7. Summary of All Key Findings

Below is a consolidated summary of every significant finding from this analysis:

- The overall portfolio default rate is **6.68%**, representing 10,025 defaulters out of 149,986 borrowers.
- Borrowers aged 21-30 have the highest default rate at **11.56%** — 1.7x the overall average. Risk falls consistently with age.
- Low income borrowers (<$3K/month) default at **9.43%**, more than double the 4.32% rate seen in high income borrowers.
- Credit utilization is the strongest predictor of default. Borrowers with utilization above 80% default at **21.08%** — nearly 9.5x the rate of low-utilization borrowers.
- The highest-risk combined segment is low income + very high utilization, with a default rate of **22.56%** — 3.38x the portfolio average.
- High debt ratio (above 0.5) is associated with significantly elevated default rates (**9.83%**), nearly 1.5x the overall rate.
- The rule-based risk score successfully validated the analysis. Score 5 borrowers default at **57.74%** — 8.64 times the average.
- **239 borrowers** had a maximum risk score of 5, and 138 of them (57.74%) actually defaulted — confirming the score identifies real high-risk individuals.

8. Recommendations

Based on the findings, below are three concrete recommendations that a risk team could realistically act on. They are derived directly from the patterns identified in the data.

Recommendation 1: Apply Enhanced Scrutiny for High Utilization Applicants

Credit utilization above 80% is the single strongest predictor of default in this dataset, with a 21.08% default rate — nearly 9.5x that of low-utilization borrowers. Lenders should flag any applicant with a utilization above 80% for additional verification steps before loan approval. This could include requesting recent bank statements, proof of income, or requiring a co-signer. This filter alone would have identified 25,131 borrowers in this dataset as high-risk.

Recommendation 2: Differentiate Risk Pricing for Young, Low-Income Borrowers

Borrowers aged 21-30 with monthly income below $3,000 represent a double-risk segment — they have the highest default rate by age (11.56%) and the highest by income (9.43%). Rather than denying these applicants outright, lenders could offer smaller loan amounts, shorter repayment windows, or slightly higher interest rates that reflect the actual risk being taken. This approach is both commercially viable and fairer to the borrower, since a total denial leaves them with no credit-building opportunity.

Recommendation 3: Use the Risk Score as an Early Warning Trigger

The five-point risk score built in this analysis (based on utilization, debt ratio, late payment history, and income level) showed a near-perfect gradient from 1.74% default at score 0 to 57.74% at score 5. Banks could implement a similar score on their existing customer base and use it as an early warning system. Customers who move from score 2 to score 3 during the loan tenure (perhaps because their utilization increased or they missed a payment) could be contacted proactively by a relationship manager — offering restructuring options before the loan goes into formal default. This is significantly cheaper for the institution than recovering a defaulted loan.

10. Project Structure & Tools Used

Tool / File	Purpose
Python 3 + Jupyter Notebook	All data cleaning, feature engineering, analysis, and visualization
Pandas	Data loading, cleaning, aggregation, and segment analysis
Seaborn + Matplotlib	All charts and visualizations
MySQL (local)	Storing cleaned data and running SQL queries
SQLAlchemy	Connecting Python to MySQL for data loading and querying
credit_risk_raw.csv	Original dataset downloaded from Kaggle
credit_risk_cleaned.csv	Cleaned and feature-engineered dataset saved after processing